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$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

where x, y and z are components of $\vec{r}$, Figure 2.25 shows the position vector $\vec{r}$.

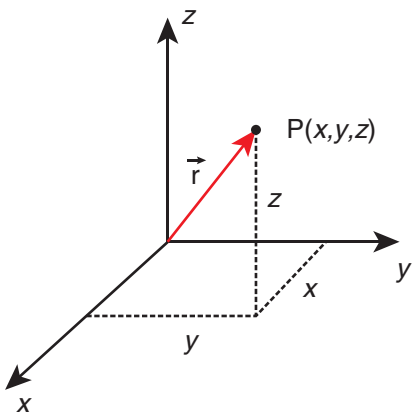
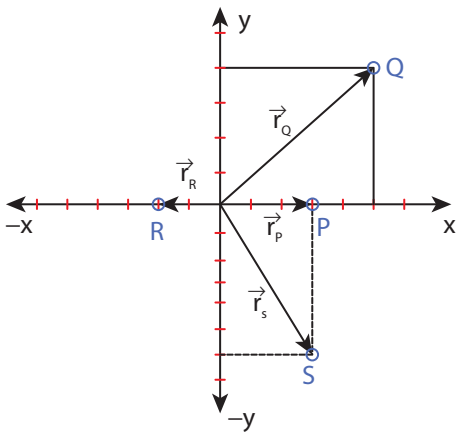


Figure 2.25 Position vector in Cartesian coordinate system

EXAMPLE 2.13

Determine the position vectors for the following particles which are located at points P, Q, R, S.



Solution

The position vector for the point P is

$$\vec{r}_P = 3\hat{i}$$

The position vector for the point Q is

$$\vec{r}_Q = 5\hat{i} + 4\hat{j}$$

The position vector for the point R is

$$\vec{r}_R = -2\hat{i}$$

The position vector for the point S is

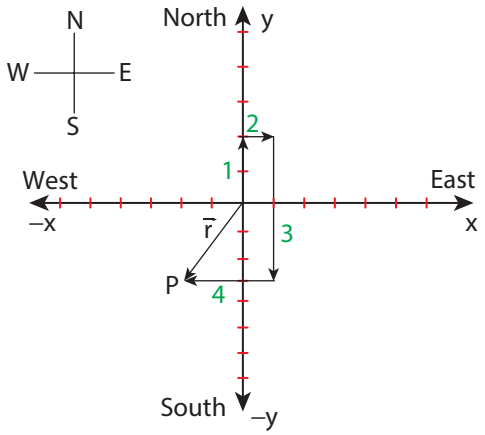
$$\vec{r}_s = 3\hat{i} - 6\hat{j}$$

EXAMPLE 2.14

A person initially at rest starts to walk 2 m towards north, then 1 m towards east, then 5 m towards south and then 3 m towards west. What is the position vector of the person at the end of the trip?

Solution

As shown in the Figure, the positive x axis is taken as east direction, positive y direction is taken as north.



After the trip, the person reaches the point P whose position vector given by

$$\vec{r} = -2\hat{i} - 3\hat{j}$$

The displacement direction is south west.